



ST BENEDICT'S

SUBJECT	P2 Marking Guidelines
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Ntombela
DATE	11 November 2024
MARKS	135
DURATION	2 hours

%

LEARNER NAME: **Marking Guidelines**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 12 pages and a separate Data Sheet and Periodic Table. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae sheet whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. A Blank page is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	Total
Mark	FM	SG	KH	FM		ZN	
Moderated Mark							
Question Total	22	26	25	15	13	34	135

Comment:

Learner's Signature:

Assessor's Signature:

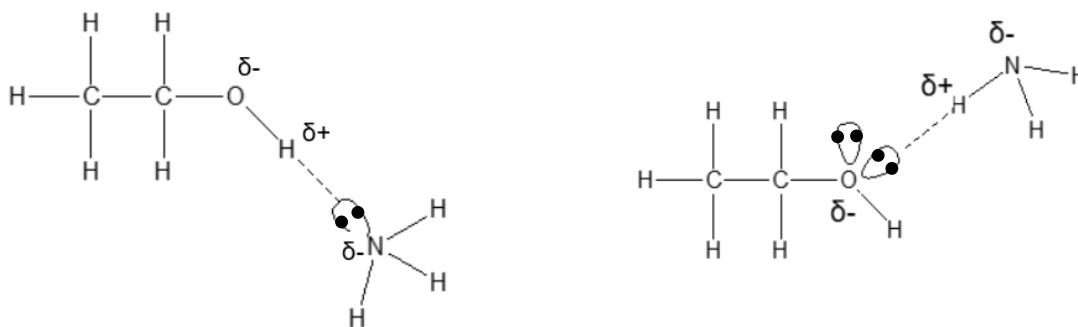
QUESTION 1

MCCOY

22 MARKS

1.1

(4)



- ✓ lone pairs and ✓ partial charges (δ^- , δ^+ , δ^-) on atoms involved in the hydrogen bond
- ✓ dotted line between lone pair on N/O to correct H
- ✓ linear O–H....N / linear N–H...O

1.2

1.2.1 A measure ✓ of the tendency of an atom to attract a bonding pair of electrons ✓. (2)

1.2.2 H and O ✓ OR O–H (1)

1.2.3 The molecule is (completely) symmetrical ✓ due to the tetrahedral shape ✓ (there is an even distribution of electron density) (2)

1.2.4 CBr₄ has London forces (LF) between molecules ✓

CHBr₃ has LF and dipole-dipole intermolecular forces ✓

Because CBr₄ has a larger mass/more electrons/larger electron cloud ✓

The LF between CBr₄ molecules are **stronger** than the dipole-dipole and London forces between CHBr₃, ✓

More energy is needed to overcome the stronger intermolecular forces between CBr₄ ✓ (5)

1.2.5 Graphite ✓ (1)

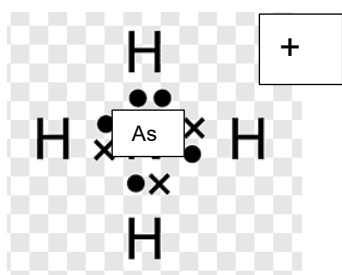
1.2.6 Graphite has delocalised electrons due to only 3 out of the 4 carbon-carbon bonds being utilised ✓.

theses electrons are free to move ✓ (when a potential difference is applied) (2)

1.3 .

1.3.1 Polar ✓ covalent ✓

(2)



1.3.2

Hydrogen has one electron ✓,

As has 5 e- ✓

lone pair and +charge ✓

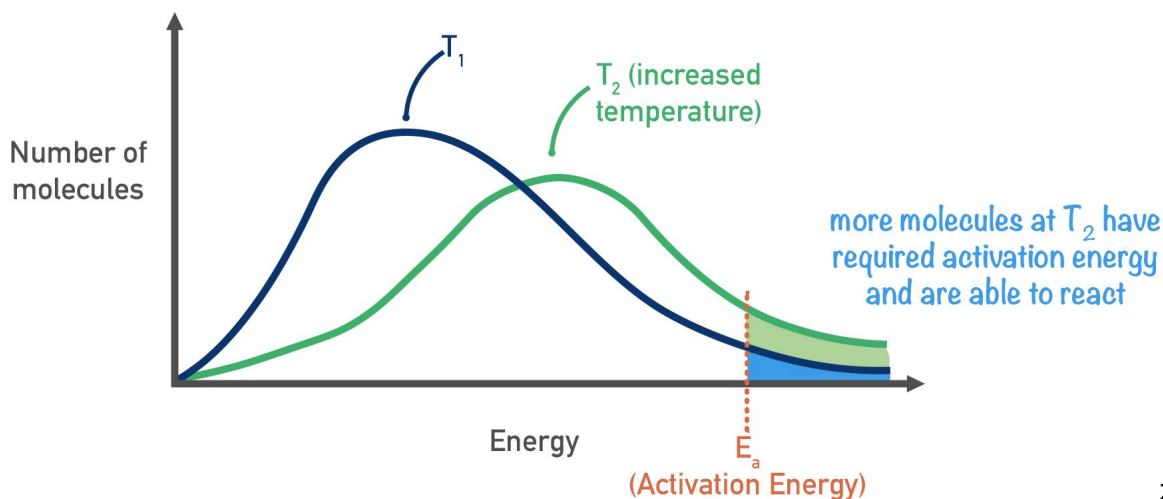
(3)

QUESTION 2

GOVENDER

26 MARKS

- 2.1 The change in the amount ✓ or concentration per unit time of either the reactant or product. ✓ (2)
- 2.2 An **increase/decrease** in concentration means that there are **more/less** particles per unit volume ✓
 There will be **more/less** collisions per unit time ✓
 There will be **more/less** effective collisions per unit time ✓
The reaction rate increases/decrease ✓ (4)
- 2.3 More particles ✓ have kinetic energy higher than or equal to the activation energy ✓ (2)
- 2.4 Axes correct ✓ (3)
 Correct curves ✓
 E_a , for higher temp is larger ✓



Z

- 2.5 same surface area of magnesium ✓
 Maintain a constant temperature for each reaction ✓
 (**NOT the same mass, not volume of HCl**) (2)
- 2.6 The 2 mol.dm^{-3} HCl solution ✓ shows the steepest gradient ✓. (2)
- 2.7 A graph starting at zero, between the 1 and 2 mol.dm^{-3} ✓
 AND ending at the same level. ✓ (2)
- 2.8 A catalyst is a substance that increase the rate of the reaction but remains unchanged by the end of the reaction. ✓
 The rate is increased by providing an alternative reaction pathway. ✓ (2)

- 2.9 A line with an even steeper ✓ initial gradient than the 2 mol.dm^{-3} solution, but half the volume of H_2 production ✓. (2)
- 2.10 Reaction rate is directly proportional ✓ to the number of effective collisions per unit time ✓, as concentration decreases, fewer particles are available for effective collisions ✓, leading to a slower reaction rate. (3)
- 2.11 Hydrogen is a non-polar molecule ✓, so it is not significantly soluble in water, allowing it to displace water effectively. ✓ (2)

QUESTION 3**HILDER****25 MARKS**

3.1 . (4)

3.1.1 NaOH; $n = cv$

$$= 2 \times 0,2$$

$$= 0,4 \text{ moles available} \quad \checkmark$$
H₂O: $n = m/M$

$$= 12/18$$

$$= 0,67 \text{ moles available} \quad \checkmark$$
NaOH: H₂O2 : 3 $\checkmark$

0,4 : 0,6 Enough

0,44 : 0,67 Not enough

0,67 moles of H₂O available, but only 0,6 required $\checkmark$. Therefore NaOH is limiting $\checkmark$

3.1.2 measure of the extent of a reaction $\checkmark$ generally measured by comparing the amount of product against the amount of product that is possible $\checkmark$ (2)

3.1.3 Theoretical yield of NaAl(OH)₄ : NaOH: NaAl(OH)₄ (5)

1: 1 $\checkmark$ **ratio**0,4 moles of NaAl(OH)₄ producedMass of NaAl(OH)₄: $m = n \times M$

$$= 0,4 \times 118 \quad \checkmark \text{Skill}$$

$$= 47,2 \quad \checkmark \text{g}$$

$$\% \text{ yield} = \text{actual} / \text{theo} \times 100 \quad \checkmark \text{eq}$$

$$= 45/47,2 \times 100$$

$$= 95,33\% \quad \checkmark$$

3.2

$$\begin{aligned}
 3.2.1 \quad \text{Fe: } n &= m/M & \text{Fe: Fe}_2\text{O}_3 \\
 &= 1000 \checkmark \text{skill} / 56 & 2: 1 \\
 &= 17,86 \text{ mols produced } \checkmark & \mathbf{8,93 \text{ moles of Fe}_2\text{O}_3 \text{ required } \checkmark} \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 3.2.2 \quad N_{\text{Fe}_2\text{O}_3} &= n \times N_a & \text{Fe: Fe}_2\text{O}_3 \\
 &= \mathbf{8,93 \text{ c.o.e.}} \times 6,02 \times 10^{23} \checkmark \text{ for } N_a & 2: 1 \checkmark \\
 &= 5,38 \times 10^{24} \text{ Iron oxide units } \checkmark \\
 &2 \times 5,38 \times 10^{24} \\
 &= \mathbf{1,075 \times 10^{25} \text{ Fe atoms } \checkmark} \quad (4)
 \end{aligned}$$

$$\begin{aligned}
 3.2.3 \quad \text{moles of CO: } &26,79 \text{ moles } \checkmark \\
 V &= nV_m \\
 &= 26,79 \times 22,4 \checkmark \text{skill} \\
 &= \mathbf{600 \text{ dm}^3 \checkmark \text{ unit and value}} \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 3.2.4 \quad \text{original mass of Fe}_2\text{O}_3 \text{ required : } m &= nM & (4) \\
 &= 8,93 \times 160 \checkmark \\
 &= 1428,8 \text{ g } \checkmark \\
 \text{Mass} \times 70\% &= 1428,8 \checkmark \\
 \mathbf{\text{Mass} = 2041,14 \text{ g of Fe}_2\text{O}_3 \text{ required } \checkmark}
 \end{aligned}$$

OR $8,93 / 0.7 = 12,75$
moles of Fe_2O_3

Mass: $m = nM$

$= 12,75 \times 160$

$= 2041,14 \text{ g of Fe}_2\text{O}_3$

QUESTION 4**MCCOY****15 MARKS**

4.1 .

4.1.1 NEGATIVE✓ (1)

4.1.2 $\text{Au} \rightarrow \text{Au}^{3+} + 3\text{e}^-$ (2)

4.1.3 a substance that can conduct electricity due to the presence of free (mobile) ions✓ when molten or dissolved in solution ✓ (2)

4.1.4 $\text{Au}(\text{NO}_3)_3$, AuCl_3 , AuBr_3 , AuCN , NaAuCl_4 , $\text{KAu}(\text{CN})_2$
✓gold ion✓balanced anion (2)

4.2 .

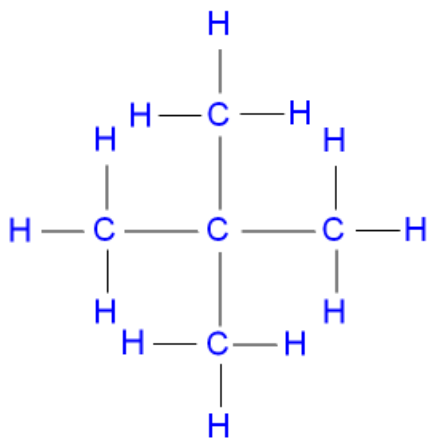
4.2.1 $\text{Cu}(\text{NO}_3)_2(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{NO}_3^-(\text{aq})$ state symbols ✓ (2)4.2.2 Ni is oxidised✓: $\text{Ni}^0 \rightarrow \text{Ni}^{2+}$ ✓, increase in oxidation number
 Cu^{2+} is reduced✓: $\text{Cu}^{2+} \rightarrow \text{Cu}^0$ ✓, decrease in oxidation number (4)4.2.3 $\text{Cu}^{2+} + \text{Ni} \rightarrow \text{Ni}^{2+} + \text{Cu}$ Reactants✓ products✓ (2)

QUESTION 5**GOVENDER****29 MARKS**

5.1 Add bromine water/solution to equal volumes of each solution in separate test tubes✓

Pent-1-ene will decolourises more quickly✓ from reddish brown to colourless✓ (3)

5.2 . ✓✓



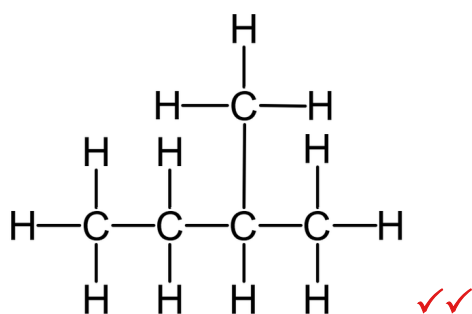
(2)

5.3 .

5.3.1 Compounds having the same molecular formula✓ but different structural formulae ✓ (2)

HILDER

5.3.2 Drawn 2-methylbutane



(2)

5.4 Pentan-1-ol and butanoic acid both have hydrogen bonds between their molecules.

✓

Butanoic acid can form two hydrogen bonds ✓ per molecule with an adjacent molecule whereas pentan-1-ol can only form one ✓.

More energy is required to overcome the greater number of hydrogen bonds ✓ between butanoic acid molecules.

(4)

QUESTION 6

NTOMBELA

34 MARKS

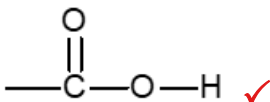
6.1 A series of similar compounds which have the same functional group and have the same general formula ✓, in which each member differs from the **previous** one by a single CH₂ unit. ✓ (2)

6.2 Haloalkanes ✓ (1)

6.3 An atom or a group of atoms ✓ that form the centre of chemical activity in the molecule ✓ (2)

6.4 .

6.4.1 C_nH_{2n+2}O₂ ✓ (1)

6.4.2  ✓ (1)

6.5 .

6.5.1 Dehydration ✓ (1)

6.5.2 Hydrolysis ✓ (1)

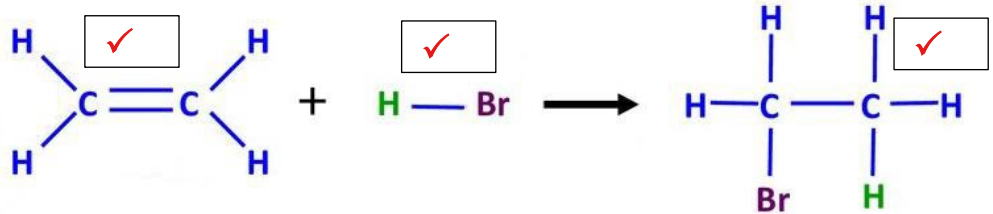
6.6 Heating of alcohol ✓ with an excess of concentrated H₂SO₄ or H₃PO₄ ✓ (2)

6.7 .

6.7.1 Water ✓ (1)

6.7.2 H₂SO₄ or H₃PO₄ ✓ (1)

6.8 .

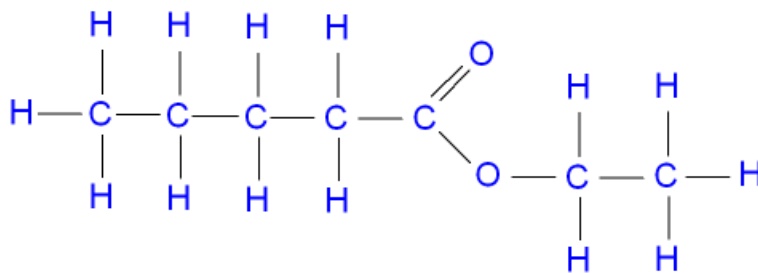
6.8.1  (3)

6.8.2 To avoid the formation of the hydroxyl group or an alcohol ✓ (1)

6.9 .

6.9.1 Esterification ✓ (1)

6.9.2 H₂SO₄ ✓ (1)



6.9.3 ✓ ✓ (2)

6.9.4 Ethyl ✓ pentanoate ✓ (2)

6.10 $\text{CH}_3\text{CH}_2\text{Br} \checkmark + \text{NaOH} \rightarrow \text{CH}_2\text{CH}_2 \checkmark + [\text{NaBr} + \text{H}_2\text{O}] \checkmark$

OR

$\text{CH}_3\text{CH}_2\text{Br} \checkmark \rightarrow \text{CH}_2\text{CH}_2 \checkmark + \text{HBr} \checkmark$ (3)

6.11 .

6.11.1 Hexane ✓
(or any branched alkane with formula C_6H_{14} e.g: 2-methylpentane) (1)

6.11.2 In shorter chained alkenes and branched alkanes the surface area is less / the IMF are weaker / the activation energy is less ✓ and they are thus more flammable ✓. (2)

6.11.3 .

6.11.4 Reaction A (1)

6.11.5 To allow for extended heating and maintains reaction concentration (2)